HVLP Application Note

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1. Connecting HVIL in a system

High Voltage Interlock Loop (HVIL) is a loop that is used to prevent high voltage from being present on the system in the event of a fault or if a connection is unplugged. Thus, if this loop opens the high voltage will be switched off.

It is expected that the high voltage supply is switched off by an external system (e.g the BMS) as normally the HVLP will not have control over this.

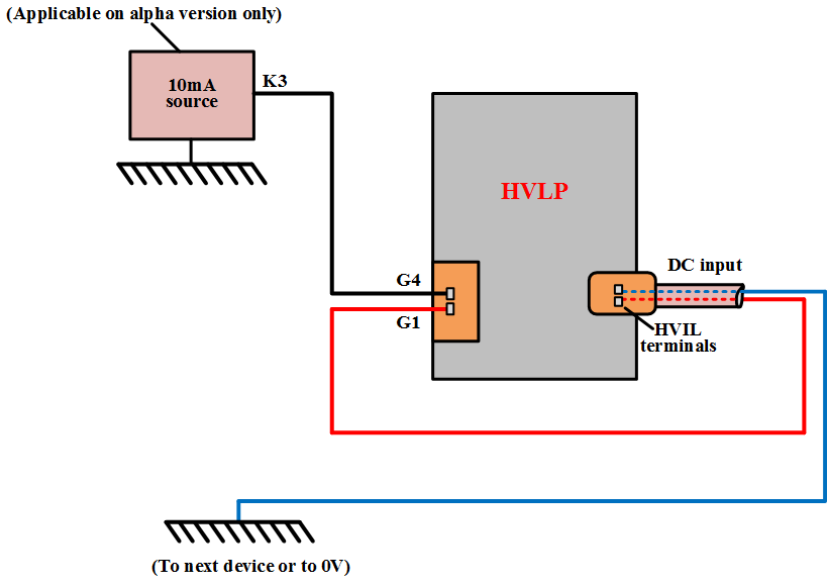


Figure 1 - Representative HVIL configuration.

NOTE: DO NOT CONNECT THE HVIL TO THE DC LINK +VE or –VE

The HVLP-10/20 alpha controller provides the following HVIL functions:

- HVIL current source which has a limited isolated supply output of 10mA maximum. It can be used to originate the interlock system. This current source is available only on HVLP alpha.
- HVIL I/O connector Inputs. Two inputs are provided on the I/O connector which can be connected either way round.
- HVIL passive loop on high voltage DC input connector. If connected a part of the HVIL, the loop will be opened if the DC input connector in unplugged.

2. Introduction to SIL

The Switch Interlock (SIL) allows to configure additional inputs available in the controller to have an Interlock feature. Additionally, SIL allows the user to configure the cutback factor to be applied when the loop is broken.

The following table provides a list of the inputs that can be used with SIL:

Pin Name	Customer Connector Pin	SIL Pin Identifier
DIN_1	J3	0
DIN_2	H3	1
DIN_3	J4	2
DIN_4	H4	3
AIN1_A	K1	4
AIN1_B	K2	5
AIN2_A	J1	6
AIN2_B	J2	7

Table 1 –SIL input table.

NOTE: For more information on how to set up general input settings such as analogue input voltage ranges, digital input polarity, etc. refer to the HVLP Product Manual.

3. HVIL/SIL Properties and Setup

Object 0x3F02 is used to enable and configure HVIL and SIL on the controller. The following table provides a list of all settings available:

Object	Function
0x3F02, 1	Enables HVIL feature on the controller.
0x3F02, 2	HVIL Loop Broken Indicator. Indicates the loop status of HVIL and SIL: 1 = loop opened (broken) 0 = loop closed
0x3F02, 3	Enables SIL feature on the controller.
0x3F02, 4	Select input used as Switch Interlock
0x3F02, 5	Cutback factor if Switch Interlock is active (0.0 – 1.0 range)

Table 2 – HVIL/SIL Properties

If either of the Interlocks is broken, there will be an Interlock loop broken fault and the cutback code reported in H-protocol will be 10 (assuming interlock is the highest priority cutback). Additionally, the HVIL Loop Broken Indicator object will report the loop as broken.

3.1. HVIL Setup

To enable HVIL in the controller set the object 'Enable HVIL feature' (0x3F02, 1) to 1.

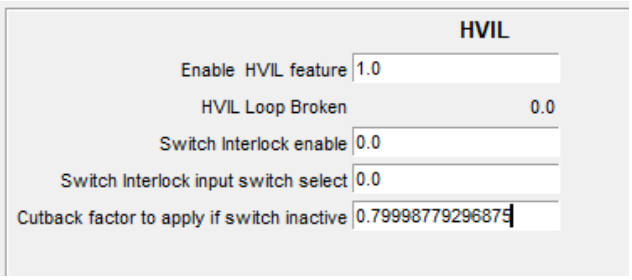


Figure 2 - HVIL DVT settings.

According to the above example, if the controller detects the HVIL loop as open, a torque cutback of 0.8 will be applied. Additionally, an "Interlock loop broken" fault (0x47C6) will be set. The fault will be cleared and the torque output will be resumed when the loop has been closed.

3.2. SIL Setup

To enable SIL on the controller:

1. Enable SIL by setting object 0x3F02 sub index 3 to 1.
2. Select the desired input to be used as SIL by setting the 0x3F02 sub index 4. For more information on the possible values to set in this object refer to table 1 (SIL Pin Identifier column).

NOTE: In order to use the analogue inputs you need to configure the object 0x4690 according to the applications needs e.g. as shown in figure below.

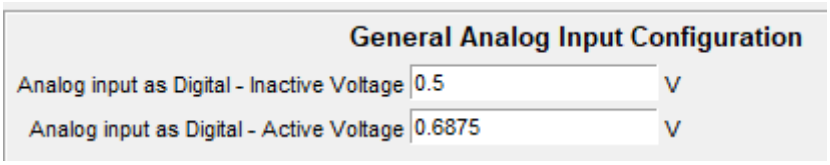


Figure 2 – Analogue input configuration DVT settings.

3. Specify the cutback factor to be applied when the loop is broken (range between 0.0 and 1.0) in object 0x3F02 sub index 5.

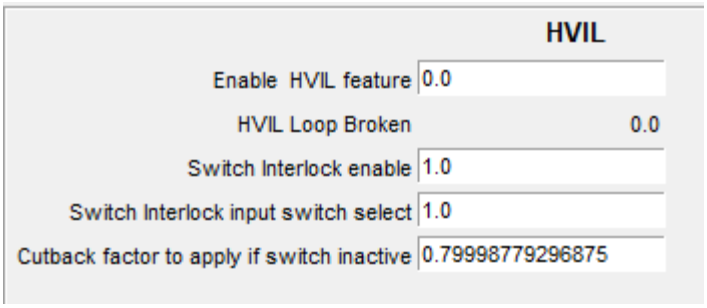


Figure 3 - SIL DVT settings.

According to the above example, when the controller detects the SIL loop as broken/open, a torque cutback of 0.8 will be applied. Additionally, an “Interlock loop broken” fault (0x47C6) will be set. The fault will be cleared and the torque output will be resumed when the loop has been closed.

NOTE: In order for the controller to work as expected, only one of the features (either HVIL or SIL) can be enabled at a time.